PRACTICE SHEET

1. $\lim_{x\to 0}\frac{\sin|x|}{x}$ equal to ?

(a) 1 (b) −1
(c) ∞ (d) Limit D.N.E.

2. $\lim_{x\to 0}\frac{x\sin 5x}{\sin^2 4x}$ equal to ?

(a) 0 (b) 5/4
(c) 5/16 (d) 25/4

3. $\lim_{x\to\infty}\left(\frac{x-2}{x+2}\right)^{x+2}$ equal to ?

(a) 0 (b) e^4
(c) e^{-2} (d) e^{-4}

4. $\lim_{x\to\infty}\frac{\sin x}{x}$ equal to ?

(a) 1 (b) 0
(c) ∞ (d) −1

5. $\lim_{x\to 0}\frac{a^x-b^x}{x}$ equal to ?

(a) $\log\left(\frac{a}{b}\right)$ (b) $\log\left(\frac{b}{a}\right)$
(c) ab (d) log(ab)

6. $\lim_{x\to\infty}\sec\theta-\tan\theta$ is equal to

(a) 0 (b) 1
(c) −1 (d) 2

7. $\lim_{x\to 0}\frac{2^x-1}{\sqrt{1+x}-1}$ equal to ?

(a) log2 (b) 2log2
(c) $\frac{1}{2}\log 2$ (d) 0

8. $\lim_{x\to\infty}x\cos\left(\frac{\pi}{4x}\right)\sin\left(\frac{\pi}{4x}\right)$ equal to ?

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$
(c) π (d) 0

9. $\lim_{x\to\infty}\left[x-\sqrt{x^2-x}\right]$ equal to ?

(a) 1/2 (b) 1
(c) −1/2 (d) 0

10. $\lim_{x\to 0}\frac{x\cos x-\sin x}{x^2\sin x}$ equal to ?

(a) 1/2 (b) −1/2
(c) 1/3 (d) −1/3

11. What is $\lim_{x\to\infty}\left(\frac{x}{3+x}\right)^{3x}$ equal to?

(a) e (b) e^3
(c) e^{-9} (d) e^9

12. What is $\lim_{x\to 0}\frac{\sin^2 ax}{bx}$ (a, b are constants) equal to?

(a) 0 (b) a
(c) $\frac{a^2}{b}$ (d) Does not exist

13. $\lim_{x\to 0}e^{-1/x}$ is equal to:

(a) 0 (b) ∞
(c) e (d) Does not exit

14. If $f(x)=\frac{1}{1-|1-x|}$, then what is $\lim_{x\to 1}f(x)$ equal to?

(a) 0 (b) ∞
(c) 1 (d) −1

15. What is the value of $\lim_{x\to\alpha}\frac{\sqrt{\alpha+2x}-\sqrt{3x}}{\sqrt{3\alpha+x}-2\sqrt{x}}$?

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{1}{(3\sqrt{3})}$
(c) $\frac{2}{(3\sqrt{3})}$ (d) $\frac{1}{\sqrt{3}}$

16. What is $\lim_{x\to 0}\frac{a^x-b^x}{x}$?

(a) $\log\left(\frac{a}{b}\right)$ (b) $\log\frac{b}{a}$
(c) ab (d) log(ab)

17. What is the value of $\lim_{x\to\infty}\left(\frac{x+6}{x+1}\right)^{x+4}$?

(a) e (b) e^2
(c) e^4 (d) e^5

18. What is the value of $\lim_{x\to 1}\frac{(x-1)^2}{|x-1|}$?

(a) 0 (b) 1
(c) −1 (d) Does not exist

ANSWER KEY

1.	d	**2.**	c	**3.**	d	**4.**	b	**5.**	a	**6.**	a	**7.**	b	**8.**	a	**9.**	a	**10.**	d
11.	c	**12.**	a	**13.**	d	**14.**	c	**15.**	c	**16.**	a	**17.**	d	**18.**	a				

Solutions

Sol.1. (d)

$\lim_{x\to 0}\frac{\sin|x|}{x}$, L.H.L is limit when x < 0

LHL= $\lim_{x\to 0}\frac{\sin(x)}{x} = -\lim_{x\to 0}\frac{\sin x}{x} = -1$

RHL = $\lim_{x\to 0}\frac{\sin(x)}{x} = 1$

So, LHL ≠ RHL

Hence, limit it does not exist

Sol.2. (c)

$$\lim_{x\to 0}\frac{x\sin 5x}{\sin^2 4x}$$

[Multiply denominator and number with x]

$$\lim_{x\to 0}\frac{x^2\sin 5x}{x\sin^2 4x} = \lim_{x\to 0}\frac{\sin 5x}{x\to 0}.\frac{x^2}{\sin^2 4x}$$

Rearranging to bring a standard form, we get,

$$\lim_{x\to 0}\frac{5\sin 5x}{5x}.\frac{(4x)^2}{16\sin^2 4x}$$

$$=\frac{5}{16}\left(\lim_{x\to 0}\frac{\sin 5x}{5x}\right).\frac{1}{\lim_{x\to 0}\left(\frac{\sin 4x}{4x}\right)^2} = \frac{5}{16}$$

Sol.3. (d)

$$\lim_{x\to 0}\left(\frac{x-2}{x+2}\right)^{x+2}$$

can be written as

$$\lim_{x\to 0}\left(\frac{x+2-4}{x+2}\right)^{x+2}$$

or $\lim_{x\to 0}\left\{1-\frac{4}{(x+2)}\right\}^{x+2}$

putting x + 2t

when x→∞, t→∞

so, $\lim_{x\to 0}\left\{1-\frac{4}{t}\right\}^{t}$

or $\lim_{x\to 0}\left\{1-\frac{4}{t}\right\}^{\frac{t}{4}\times 4}$

or, $\lim_{x\to 0}\left\{\left(1-\frac{4}{t}\right)^{\frac{t}{4}}\right\}^4 = (e^{-1})^4 = e^{-4}$

Sol.4. (b)

Given limit is : $\lim_{x\to 0}\frac{\sin}{x}$

Let, x=1/h and as x→∞, h→0

∴ $\lim_{x\to 0}\frac{\sin x}{x}$ change to : $\lim_{h\to 0} h \sin\frac{1}{h}$

=0 (value of sin 1/h is finite as it lies between–1 and 1)

Sol.5. (a)

Let L= $\lim_{x\to 0}\frac{a^x - b^x}{x}$

This is of 0/0 form so, L' hospital's rule is applicable

⇒L= $\lim_{x\to 0}\frac{a^x - \log a - b^x \log b}{1}$

(by L' Hospital's rule)

=log a – log b = log a/b

Sol. 6. (a)

$$\lim_{\theta\to\frac{\pi}{2}}\sec\theta-\tan\theta = \lim_{\theta\to\frac{\pi}{2}}\frac{1-\sin\theta}{\cos\theta}$$

apply L hospital

$$\lim_{\theta\to\frac{\pi}{2}}\frac{-\cos\theta}{-\sin\theta} = 0$$

Sol. 7. (b)

$$\lim_{x\to 0}\frac{2^x-1}{\sqrt{1+x}-1}$$

apply L hospital

$$\lim_{x\to 0}\frac{2^x\log 2}{1\Big/2\sqrt{1+x}} = 2\log 2$$

Sol. 8. (a)

$$\lim_{x\to\infty} x\cos\left(\frac{\pi}{4x}\right)\sin\left(\frac{\pi}{4x}\right)$$

$$\lim_{x\to\infty}\frac{x}{2}\sin\left(\frac{\pi}{2x}\right) = \lim_{x\to\infty}\frac{\sin\left(\frac{\pi}{2x}\right)}{\frac{2}{x}}$$

$$=\lim_{x\to\infty}\frac{\sin\left(\frac{\pi}{2x}\right)}{\frac{\pi}{2x}}\frac{\pi}{4} = \frac{\pi}{4}$$

Sol. 9. (a)

$$\lim_{x\to\infty}\left[x-\sqrt{x^2-x}\right]$$

$$=\lim_{x\to\infty}\left[x-\sqrt{x^2-x}\right]\times\frac{x+\sqrt{x^2-x}}{x+\sqrt{x^2-x}}$$

$$=\lim_{x\to\infty}\frac{x}{x+\sqrt{x^2-x}} = \frac{1}{2}$$

Sol. 10. (d)

$$\lim_{x\to 0}\frac{x\cos x-\sin x}{x^2\sin x}\quad\left(\frac{0}{0}\text{form}\right)$$

apply L hospital

$$\lim_{x\to 0}\frac{\cos x - x\sin x-\cos x}{x^2\cos x+2x\sin x} = \lim_{x\to 0}\frac{-\sin x}{x\cos x+2\sin x}$$

again apply L hospital

$$=\lim_{x\to 0}\frac{-\cos x}{\cos x - x\sin x+2\cos x} = -\frac{1}{3}$$

Sol. 11. (c)

$$\lim_{x\to\infty}\left(\frac{x}{3+x}\right)^{3x}$$

$$=\lim_{x\to\infty}\left(\frac{x+3-3}{3+x}\right)^{3x} = \lim_{x\to\infty}\left(1+\frac{-3}{3+x}\right)^{3x} = e^{-9}$$

Sol. 12. (a)

$$\lim_{x\to 0}\frac{\sin^2 ax}{bx}$$

$$\lim_{x\to 0}\frac{\sin^2 ax}{a^2x^2}\times\frac{a^2x}{b} = 1\times\frac{a^2(0)}{b} = 0$$

Sol. 13. (d)

$\lim_{x\to 0} e^{-1/x}$

L.H.L.

$$\lim_{h\to 0} e^{-1/(0-h)} = \lim_{h\to 0} e^{1/h} = e^{\infty} = \infty$$

R.H.L.

$$\lim_{h\to 0} e^{-1/(0+h)} = e^{-\infty} = 0$$

L.H.L. ≠ R.H.L so limit does not exists.

Sol. 14. (c)

$$\lim_{x\to 1}\frac{1}{1-|1-x|} = 1$$

Sol. 15. (c)

$$\lim_{x\to\alpha}\frac{\sqrt{\alpha+2x}-\sqrt{3x}}{\sqrt{3\alpha+x}-2\sqrt{x}}$$

rationalize the numerator and denominator

$$\lim_{x\to\alpha}\frac{\sqrt{\alpha+2x}-\sqrt{3x}}{\sqrt{3\alpha+x}-2\sqrt{x}}\times\frac{\sqrt{3\alpha+x}+2\sqrt{x}}{\sqrt{3\alpha+x}+2\sqrt{x}}\times\frac{\sqrt{\alpha+2x}+\sqrt{3x}}{\sqrt{\alpha+2x}+\sqrt{3x}}$$

$$=\lim_{x\to\alpha}\frac{\alpha+2x-3x}{3\alpha+x-4x}\times\frac{\sqrt{3\alpha+x}+2\sqrt{x}}{\sqrt{\alpha+2x}+\sqrt{3x}}$$

$$\lim_{x\to\alpha}\frac{\alpha-x}{3(\alpha-x)}\times\frac{\sqrt{3\alpha+x}+2\sqrt{x}}{\sqrt{\alpha+2x}+\sqrt{3x}} = \frac{4}{6\sqrt{3}} = \frac{2}{3\sqrt{3}}$$

Sol. 16. (a)

$$\lim_{x\to 0}\frac{a^x-b^x}{x}?$$

apply L. Hospital rule

$$\lim_{x\to 0}\frac{a^x\log a-b^x\log b}{1} = \log a-\log b = \log\left(\frac{a}{b}\right)$$

Sol. 17. (d)

$$\lim_{x\to\infty}\left(\frac{x+6}{x+1}\right)^{x+4}$$

$$\lim_{x\to\infty}\left(\frac{x+1+5}{x+1}\right)^{x+4} = \lim_{x\to\infty}\left(1+\frac{5}{x+1}\right)^{x+4}$$

by standard limit

$= e^5$

Sol. 18. (a)

$$\lim_{x\to 1}\frac{(x-1)^2}{|x-1|}?$$

$$\lim_{x\to 1}\frac{|x-1|^2}{|x-1|} = \lim_{x\to 1}|x-1| = 0$$

OS. OFFICIUM SUPREMUM

NDA PYQ

1. What is $\lim_{x\to\infty}\left(\sqrt{a^2x^2+ax+1}-\sqrt{a^2x^2+1}\right)$ equal to?

(a) 1/2 (b) 1
(c) 2 (d) 0

[NDA (I) - 2011]

2. What is the value of $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$?

(a) 0 (b) 1
(c) 1/2 (d) limit does not exist

[NDA (I) - 2012]

3. What is the value of $\lim_{x\to -2}\left(\frac{x+2}{x^3+8}\right)$?

(a) 1/4 (b) –1/4
(c) 1/12 (d) –1/12

[NDA (I) - 2012]

4. What is the value of $\lim_{x\to 0}\frac{2(1-\cos x)}{x^2}$?

(a) 0 (b) 1/2
(c) 1/4 (d) 1

[NDA (II) - 2012]

5. What is the value of $\lim_{x\to 2}\frac{x-2}{x^2-4}$?

(a) 0 (b) 1/4
(c) 1/2 (d) 1

[NDA (II) - 2012]

6. Consider the following statements:

I. $\lim_{x\to 0}\frac{1}{x}$ exists.

II. $\lim_{x\to 0} e^{1/x}$ does not exists

Which of the above statements is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (II) - 2012]

7. What is the value of $\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x}$?

(a) 0 (b) 1/2
(c) 1 (d) –1/2

[NDA (II) - 2012]

8. Consider the following statements:

I. $\lim_{x\to 0}\sin\frac{1}{x}$ does not exist

II. $\lim_{x\to 0} x\sin\frac{1}{x}$ exists

Which of the above statements is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2013]

9. Which is $\lim_{x\to 0}\frac{\sin x-\tan x}{x}$ equal to?

(a) 0 (b) 1
(c) –1 (d) 1/2

[NDA (I) - 2013]

10. What is the value of $\lim_{x\to 0}\frac{1-\sqrt{1+x}}{x}$?

(a) 0 (b) 1
(c) -1 (d) –1/2

[NDA (I) - 2013]

11. What is $\lim_{x\to 2}\frac{2-x}{x^3-8}$ equal to?

(a) 1/8 (b) –1/8
(c) 1/12 (d) –1/12

[NDA (II) - 2013]

12. What is $\lim_{x\to 0}\frac{1-\cos x}{x}$ equal to?

(a) 0 (b) 1/2
(c) 1 (d) 2

[NDA (II) - 2013]

13. What is $\lim_{x\to 0}\frac{\cos x}{\pi - x}$ equal to?

(a) 0 (b) π
(c) $1/\pi$ (d) 1

[NDA (II) - 2013]

14. What is $\lim_{x\to 0}\frac{\sin 2x+4x}{2x+\sin 4x}$ equal to?

(a) 0 (b) 1/2
(c) 1 (d) 2

[NDA (II) - 2013]

15. f(9) = 9 and f'(9)=4, then what is $\lim_{x\to 9}\frac{\sqrt{f(x)}-3}{\sqrt{x}-3}$ equal to?

(a) 36 (b) 9
(c) 4 (d) None of these

[NDA (I) - 2014]

16. What is $\lim_{x\to 0}\frac{(1+x)^n-1}{x}$ equal to?

(a) 0 (b) 1
(c) n (d) n–1

[NDA (I) - 2014]

17. What is $\lim_{x\to 0}\frac{x}{\sqrt{1-\cos x}}$ equal to?

(a) $\sqrt{2}$ (b) $-\sqrt{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) Does not exist

[NDA (I) - 2014]

18. What is $\lim_{x\to 0}\frac{\log_5(1+x)}{x}$ equal to?

(a) 1 (b) $\log_5 e$
(c) $\log_e 5$ (d) 5

[NDA (II) - 2014]

19. What is $\lim_{x\to 0}\frac{5^x-1}{x}$ equal to?

(a) $\log_e 5$ (b) $\log_5 e$
(c) 5 (d) 1

[NDA (II) - 2014]

OFFICIUM SUPREMUM

20. What is $\lim_{n\to\infty}\frac{1+2+3+......+n}{1^2+2^2+3^2+........+n^2}$ equal to?

(a) 5 (b) 2
(c) 1 (d) 0

[NDA (II) - 2014]

21. If G(x) = $\sqrt{25-x^2}$, then what is $\lim_{x\to1}\frac{G(x)-G(1)}{x-1}$ equal to?

(a) $-\frac{1}{2\sqrt{6}}$ (b) $\frac{1}{5}$

(c) $-\frac{1}{6}$ (d) $\frac{1}{\sqrt{6}}$

[NDA (I) - 2015]

Directions (for next two): Given that $\lim_{x\to\infty}\left(\frac{2+x^2}{1+x}-Ax-B\right)=3$

22. What is the value of A?

(a) –1 (b) 1
(c) 2 (d) 3

[NDA (I) - 2015]

23. What is the value of B?

(a) –1 (b) 3
(c) –4 (d) –3

[NDA (I) - 2015]

24. If f(x) = $\sqrt{25-x^2}$, then what is $\lim_{x\to1}\frac{f(x)-f(1)}{x-1}$ equal to?

(a) $\frac{1}{5}$ (b) $\frac{1}{24}$

(c) $\sqrt{24}$ (d) $-\frac{1}{\sqrt{24}}$

[NDA (II) - 2015]

25. If f(x) = $\frac{\sin(e^{x-2}-1)}{\text{In}(x-1)}$, then $\lim_{x\to2}$ (x) is equal to?

(a) –2 (b) –1
(c) 0 (d) 1

[NDA (II) - 2015]

For the two (2) items that follow:

$f(x)=\frac{a^{[x]+x}-1}{[x]+x}$

where[.] denotes the greatest integer function.

26. What is $\lim_{x\to0^+}f(x)$ equal to?

(a) 1 (b) In a
(c) 1 – a^{-1} (d) Limit does not exit

[NDA (I) - 2016]

27. What is $\lim_{x\to0^-}$ f(x) equal to?

(a) 0 (b) In a
(c) 1 – a^{-1} (d) Limit does not exist

[NDA (I) - 2016]

28. What is $\lim_{x\to0}e^{-\frac{1}{x^2}}$ equal to?

(a) 0 (b) 1
(c) –1 (d) Limit does not exist

[NDA (I) - 2016]

29. If $\lim_{x\to0}\phi$ (x) = a^2, where a ≠ 0, then what is $\lim_{x\to0}\phi\left(\frac{x}{a}\right)$ equal to?

(a) a^2 (b) a^{-2}
(c) $-a^2$ (d) –a

[NDA (I) - 2016]

30. What is $\lim_{x\to0}\frac{e^x-(1+x)}{x^2}$ equal to?

(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

[NDA (I) - 2017]

31. If F (x) = $\sqrt{9-x^2}$, then what is $\lim_{x\to1}\frac{f(x)-F(1)}{x-1}$ equal to:

(a) $-\frac{1}{4\sqrt{2}}$ (b) $\frac{1}{8}$

(c) $-\frac{1}{2\sqrt{2}}$ (d) $\frac{1}{2\sqrt{2}}$

[NDA (I) - 2017]

32. Consider the following statements:

1.If $\lim_{x\to a}f(x)$ and $\lim_{x\to a}g(x)$ both exits, then $\lim_{x\to a}$ {f(x)g (x)} exists.

2.If $\lim_{x\to a}$ {f(x) g (x)} exists, then both $\lim_{x\to a}$ f(x) and $\lim_{x\to a}$ g(x) must exits.

Which of the above statements is /are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2017]

33. If $\lim_{x\to\frac{\pi}{2}}\frac{\sin x}{x}=l$ and $\lim_{x\to\infty}\frac{\cos x}{x}$ = m, then which one of the following is correct?

(a) I =1, m = 1 (b) $I=\frac{2}{\pi}$, m = –∞

(c) $I=\frac{2}{\pi}$, m = 0 (d) I = 1, m = ∞

[NDA (II) - 2017]

34. What is $\lim_{h\to0}\frac{\sqrt{2x+3h}-\sqrt{2x}}{2h}$ equal to?

(a) $\frac{1}{2\sqrt{2x}}$ (b) $\frac{3}{\sqrt{2x}}$

(c) $\frac{3}{2\sqrt{2x}}$ (d) $\frac{3}{4\sqrt{2x}}$

[NDA (I) - 2018]

35. What is $\lim_{x\to0}\frac{\tan x}{\sin 2x}$ equal to?

(a) 1/2 (b) 1
(c) 2 (d) Limit does not exist

[NDA (I) - 2018]

36. What is $\lim_{x\to\frac{\pi}{6}}\frac{2\sin^2x+\sin x-1}{2\sin^2x-3\sin x+1}$ to?

(a) $-\frac{1}{2}$ (b) $-\frac{1}{3}$

(c) –2 (d) –3

[NDA (II) - 2018]

37. What is $\lim_{\theta\to0}\frac{\sqrt{1-\cos\theta}}{\theta}$ equal to?

(a) $\sqrt{2}$ (b) $2\sqrt{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) $-\frac{1}{2\sqrt{2}}$

[NDA (II) - 2018]

38. If f(x) = $\sqrt{25-x^2}$, then what is $\lim_{x\to1}\frac{f(x)-f(1)}{x-1}$ equal to:

(a) $-\frac{1}{\sqrt{24}}$ (b) $\frac{1}{\sqrt{24}}$
(c) $-\frac{1}{4\sqrt{3}}$ (d) $\frac{1}{4\sqrt{3}}$

[NDA (II) - 2018]

39. $\lim_{x\to0}\frac{1-\cos^3 4x}{x^2}$ equal to?

(a) 0 (b) 12
(c) 24 (d) 36

[NDA (I) - 2019]

40. What is the value of $\lim_{x\to0}\frac{\sin x^0}{\tan 3x^0}$?

(a) 1/4 (b) 1/3
(c) 1/2 (d) 1

[NDA (II) - 2019]

41. What is $\lim_{x\to0}\frac{3^x+3^{-x}-2}{x}$ equal to ?

(a) 0 (b) −1
(c) 1 (d) limit does not exist

[NDA 2020]

42. What is $\lim_{x\to1}\frac{x+x^2+x^3-3}{x-1}$ equal to ?

(a) 1 (b) 2
(c) 3 (d) 6

[NDA 2020]

43. If $\lim_{x\to1}\frac{x^4-1}{x-1}=\lim_{x\to k}\frac{x^3-k^3}{x^2-k^2}$, where k≠0 , then what is the value of k ?

(a) 2/3 (b) 4/3
(c) 8/3 (d) 4

[NDA 2020]

44. What is $\lim_{x\to0}\frac{\sin x\log(1-x)}{x^2}$ equal to?

(a) −1 (b) 0
(c) −e (d) −1/e

[NDA 2020]

45. If a differentiable function f(x) satisfies
$\lim_{x\to-1}\frac{f(x)}{x^2-1}=-\frac{3}{2}$ then what is the value of $\lim_{x\to-1}f(x)$?

(a) $-\frac{3}{2}$ (b) −1
(c) 0 (d) 1

[NDA (I) - 2021]

46. $\lim_{x\to a}\frac{a^x-x^a}{x^a-a^a}=-1$
then what is the value of a ?

(a) −1 (b) 0
(c) 1 (d) 2

[NDA (I) - 2021]

47. What is $\lim_{x\to-1}\frac{x^3+x^2}{x^2+3x+2}$ equal to ?

(a) 0 (b) 1
(c) 2 (d) 3

[NDA (I) - 2021]

48. What is $\lim_{n\to\infty}\frac{a^n+b^n}{a^n-b^n}$ where a > b > 1, equal to?

(a) –1 (b) 0
(c) 1 (d) Limits does not exist

[NDA (II) - 2021]

49. Let f(x)= $\begin{cases}1+\frac{x}{2k}, & 0<x<2\\ kx, & 2\le x<4\end{cases}$

If $\lim_{x\to2}$ f(x) exists, then what is the value of k?

(a) –2 (b) –1
(c) 0 (d) 1

[NDA (II) - 2021]

50. If f(x)= $\frac{[x]}{|x|}$, x ≠ 0, where [.] denotes the greatest integer function, then what is the right-hand limit of f(x) at x = 1?

(a) –1
(b) 0
(c) 1
(d)Right-hand limit of f(x) at x = 1 does not exist

[NDA (II) - 2021]

51. What is $\lim_{x\to0}$ x^3 (cosec)^2 equal to?

(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) Limit does not exit

[NDA (I) - 2022]

52. What is $\lim_{x\to1}\frac{x^3-1}{\sqrt{x}-1}$ equal to?

(a) 0 (b) 3
(c) 6 (d)Limit does not exist

[NDA (I) - 2022]

53. What is $\lim_{x\to0}\frac{x}{\sqrt{1-\cos 4x}}$ equal to?

(a) $\frac{1}{2\sqrt{2}}$ (b) $-\frac{1}{2\sqrt{2}}$
(c) $\sqrt{2}$ (d) Limit does not exist

[NDA 2022 (II)]

54. What is $\lim_{x\to\frac{\pi}{2}}\frac{4x-2\pi}{\cos x}$ equal to?

(a) – 4 (b) –2
(c) 2 (d) 4

[NDA 2022 (II)]

55. If f(x) = $\frac{x^2+x+|x|}{x}$, then what is $\lim_{x\to0}f(x)$ equal to?

(a) 0 (b) 1
(c) 2 (d) $\lim_{x\to0}f(x)$ does not exist

[NDA 2022 (II)]

56. What is $\lim_{h\to0}\frac{\sin^2(x+h)-\sin^2 x}{h}$ equal to?

(a) $\sin^2 x$ (b) $\cos^2 x$
(c) sin2x (d) cos2x

[NDA 2022 (II)]

57. What is $\lim_{x\to 5}\frac{5-x}{|x-5|}$ equal to?

(a) –1 (b) 0
(c) 1 (d) Limit does not exits

[NDA – 2023 (1)]

58. What is $\lim_{x\to 1}\frac{x^9-1}{|x^3-1|}$ equal to?

(a) –1 (b) –3
(c) 3 (d) Limit does not exit

[NDA – 2023 (1)]

Consider the following for the next three (03) items that follow:

Let $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$

59. What is f(0) equal to?

(a) –1 (b) 0
(c) 1 (d) 2

[NDA – 2023 (1)]

60. What is $\lim_{x\to 0}\frac{f(x)}{x}$ equal to?

(a) –1 (b) 0
(c) 1 (d) 2

[NDA – 2023 (1)]

61. What is $\lim_{x\to 0}\frac{f(x)}{x^2}$ equal to?

(a) –1 (b) 0
(c) 1 (d) 2

[NDA – 2023 (1)]

Consider the following for the next (02) items that follow

Let $f(x) = \frac{a^{x-1}+b^{x-1}}{2}$ and g(x) = x–1

62. What is $\lim_{x\to 1}\frac{f(x)-1}{g(x)}$ equal to?

(a) $\frac{\ell n(ab)}{4}$ (b) $\frac{\ell n(ab)}{2}$
(c) $ln(ab)$ (d) $2ln(ab)$

[NDA-2023 (2)]

63. What is $\lim_{x\to 1} f(x)^{\frac{1}{g(x)}}$ equal to?

(a) √ab (b) ab
(c) 2ab (d) $\frac{\sqrt{ab}}{2}$

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

Let f(x) = |x| and g(x) = [x] –1, where [.] is the greatest integer function.

Let $h(x)=\frac{f(g(x))}{g(f(x))}$

64. What is $\lim_{x\to 0+} h(x)$ equal to?

(a) –2 (b) –1
(c) 0 (d) 1

[NDA-2023 (2)]

65. What is $\lim_{x\to 0-} h(x)$ equal to?

(a) –2 (b) –1
(c) 0 (d) 2

[NDA-2023 (2)]

66. Let f(x) = |x| + 1 and g(x) = [x] – 1,. Where [.] is the greatest integer function ,

Let $h(x)=\frac{f(x)}{g(x)}$

What is $\lim_{x\to 0-} h(x)+\lim_{x\to 0+} h(x)$ eqaual to?

(a) $-\frac{3}{2}$ (b) $-\frac{1}{2}$
(c) $\frac{1}{2}$ (d) $\frac{3}{2}$

[NDA-2024 (1)]

67. Which one of the following is correct regarding $\lim_{x\to 3}\frac{|x-3|}{x-3}$?

(a) Limit exists and is equal to 1
(b) Limit exists and is equal to 0
(c) Limit exists and is equal to – 1
(d) Limit does not exists

[NDA-2024 (2)]

68. What is $\lim_{\theta\to\frac{\pi}{2}}(\sec\theta-\tan\theta)$ equal to ?

(a) – 1 (b) 0
(c) ½ (d) 1

[NDA-2024 (2)]

Consider the following for the two (02) items that follow:

Let the function f(x) = sin[x], where [·] is the greatest integer function and g(x) = |x|

69. What is $\lim_{x\to 0}\{f(x)g(x)\}$ equal to?

(a) –1 (b) 0
(c) 1 (d) Limit does not exist

[NDA-2025 (1)]

70. What is $\lim_{x\to 0}\frac{f(x)}{g(x)}$ equal to?

(a) –sin 1 (b) sin 1
(c) 0 (d) Limit does not exist

[NDA-2025 (1)]

Consider the following for the one (01) items that follow:

Let the function $f(x) = x^2 - 1$

71. What is $\lim_{x\to 1}\{fof(x)\}$ equal to?

(a) –1 (b) 0
(c) 1 (d) 2

[NDA-2025 (1)]

Consider the following for the one (01) items that follow:

Let the function $f(x) = x^2 + 9$

72. What is $\lim_{x\to 0}\frac{\sqrt{f(x)}-3}{\sqrt{f(x)+7}-4}$ equal to?

(a) 2/3 (b) 1
(c) 4/3 (d) 2

[NDA-2025 (1)]

For the following **two (02) items**:

Let

OFFICIUM SUPREMUM

$$f(x)=\begin{cases}\dfrac{1-\cos 2x}{x^2} & x<0\\ 9 & x=0\\ \dfrac{\sqrt{x}}{\sqrt{\left(16+\sqrt{x}\right)}-4} & x>0\end{cases}$$

73. What is $\lim\limits_{x\to 0^-} f(x)$ equal to?

(a) 2 (b) 4

(c) 6 (d) 8

[NDA-2025 (2)]

74. What is $\lim\limits_{x\to 0^+} f(x)$ equal to?

(a) 6 (b) 7

(c) 8 (d) 9

[NDA-2025 (2)]

For the following **one** items:

Consider the function f(x) = x|x|

75. What is $\lim\limits_{x\to -1} f(x)$ equal to?

(a) – 1 (b) 0

(c) 1 (d) Limit does not exist

[NDA-2025 (2)]

ANSWER KEY

1.	a	**2.**	a	**3.**	c	**4.**	d	**5.**	b	**6.**	b	**7.**	b	**8.**	c	**9.**	a	**10.**	d
11.	d	**12.**	a	**13.**	c	**14.**	c	**15.**	c	**16.**	c	**17.**	d	**18.**	b	**19.**	a	**20.**	d
21.	a	**22**	b	**23.**	c	**24.**	d	**25.**	d	**26.**	b	**27.**	c	**28.**	a	**29.**	a	**30.**	b
31.	c	**32.**	a	**33.**	c	**34.**	d	**35.**	a	**36.**	d	**37.**	c	**38.**	a	**39.**	c	**40.**	b
41.	a	**42.**	d	**43.**	c	**44.**	a	**45.**	c	**46.**	c	**47.**	b	**48.**	c	**49.**	d	**50.**	c
51.	a	**52.**	c	**53.**	d	**54.**	a	**55.**	d	**56.**	c	**57.**	d	**58.**	d	**59.**	b	**60.**	b
61.	a	**62.**	b	**63.**	a	**64.**	b	**65.**	a	**66.**	a	**67.**	d	**68.**	b	**69.**	b	**70.**	d
71.	a	**72.**	c	**73.**	a	**74.**	d	**75.**	a										

Solutions

Sol. 1. (a)

$$\lim_{x\to\infty}\left(\sqrt{a^2x^2+ax+1}-\sqrt{a^2x^2+1}\right)$$

rationalize

$$\lim_{x\to\infty}\left(\sqrt{a^2x^2+ax+1}-\sqrt{a^2x^2+1}\right)\times\frac{\left(\sqrt{a^2x^2+ax+1}+\sqrt{a^2x^2+1}\right)}{\left(\sqrt{a^2x^2+ax+1}+\sqrt{a^2x^2+1}\right)}$$

$$\lim_{x\to\infty}\frac{\left(a^2x^2+ax+1-a^2x^2-1\right)}{\left(\sqrt{a^2x^2+ax+1}+\sqrt{a^2x^2+1}\right)}$$

$$\lim_{x\to\infty}\frac{(ax)}{\left(\sqrt{a^2x^2+ax+1}+\sqrt{a^2x^2+1}\right)}=\frac{a}{a+a}=\frac{1}{2}$$

Sol. 2. (a)

$$\lim_{x\to 0}x^2\sin\left(\frac{1}{x}\right)$$

= 0 × (any value between −1 and 1) = 0

Sol. 3. (c)

$$\lim_{x\to -2}\left(\frac{x+2}{x^3+8}\right)$$

apply L. Hospital

$$\lim_{x\to -2}\left(\frac{1}{3x^2}\right)=\frac{1}{12}$$

Sol. 4. (d)

$$\lim_{x\to 0}\frac{2(1-\cos x)}{x^2}$$

apply L. Hospital

$$\lim_{x\to 0}\frac{2(\sin x)}{2x}=1$$

Sol. 5. (b)

$$\lim_{x\to 2}\frac{x-2}{x^2-4}$$

apply L. Hospital

$$\lim_{x\to 2}\frac{1}{2x}=\frac{1}{4}$$

Sol. 6. (b)

I. $\lim_{x\to 0}\frac{1}{x}$

R.H.L. $\lim_{h\to 0}\frac{1}{h}=\infty$

L.H.L $\lim_{h\to 0}\frac{1}{-h}=-\infty$

$\lim_{x\to 0}\frac{1}{x}$ does not exists.

II. $\lim_{x\to 0}e^{1/x}$ does not exists

R.H.L. $\lim_{h\to 0}e^{1/h}=\infty$

L.H.L $\lim_{h\to 0}e^{1/-h}=0$

$\lim_{x\to 0}\frac{1}{x}$ does not exists.

only II statement is correct.

Sol. 7. (b)

$$\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x}?$$

apply L. Hospital

$$\lim_{x\to 0}\frac{\frac{1}{2\sqrt{1+x}}}{1}=\frac{1}{2}$$

Sol. 8. (c)

$$\lim_{x\to 0}\sin\frac{1}{x}$$

its value oscillates between -1 and 1 so limit does not exist.

now $\lim_{x\to 0}x\sin\frac{1}{x}$

= 0 × (any value between -1 and 1) = 0

limit exists.

Sol. 9. (a)

$$\lim_{x\to 0}\frac{\sin x-\tan x}{x}=0$$

standard limits.

Sol. 10. (d)

$$\lim_{x\to 0}\frac{1-\sqrt{1+x}}{x}$$

apply L. Hospital

$$\lim_{x\to 0}\frac{-\frac{1}{2\sqrt{1+x}}}{1}=-\frac{1}{2}$$

Sol. 11. (d)

$$\lim_{x\to 2}\frac{2-x}{x^3-8}$$

apply L. hospital

$$\lim_{x\to 2}\frac{-1}{3x^2}=\frac{-1}{12}$$

Sol. 12. (a)

$$\lim_{x\to 0}\frac{1-\cos x}{x}=$$

apply L. hospital

$$\lim_{x\to 0}\frac{\sin x}{1}=0$$

Sol. 13. (c)

$$\lim_{x\to 0}\frac{\cos x}{\pi-x}=\frac{1}{\pi}$$

Sol. 14. (c)

$$\lim_{x\to 0}\frac{\sin 2x+4x}{2x+\sin 4x}$$

apply L. Hospital

$$\lim_{x\to 0}\frac{2\cos 2x+4}{2+4\cos 4x}=\frac{6}{6}=1$$

Sol. 15. (c)

$$\lim_{x\to 9}\frac{\sqrt{f(x)}-3}{\sqrt{x}-3}$$

apply L. hospital

$$\lim_{x\to 9}\frac{\frac{1}{2\sqrt{f(x)}}f'(x)}{\frac{1}{2\sqrt{x}}}=\lim_{x\to 9}\frac{\sqrt{x}\times f'(x)}{\sqrt{f(x)}}$$

$$=\frac{\sqrt{9}\times f'(9)}{\sqrt{f(9)}}=\frac{4\times 3}{3}=4$$

Sol. 16. (c)

$$\lim_{x\to 0}\frac{(1+x)^n-1}{x}$$

apply L. Hospital

$$\lim_{x\to 0}\frac{n(1+x)^{n-1}}{1}=n$$

Sol. 17. (d)

$$\lim_{x\to 0}\frac{x}{\sqrt{1-\cos x}}$$

$$\lim_{x\to 0}\frac{x}{\sqrt{2}\left|\sin\frac{x}{2}\right|}$$

L.H.L is negative and R.H.L. is positive for this. So limit does not exist.

Sol. 18. (c)

$$\lim_{x\to 0}\frac{\log_5(1+x)}{x}$$

apply L. Hospital

$$\lim_{x\to 0}\frac{1}{1+x}\cdot\frac{1}{\log_5 e}=\log_e 5$$

Sol. 19. (a)

$$\lim_{x\to 0}\frac{5^x-1}{x}$$

apply L. Hospital

$$\lim_{x\to 0}\frac{5^x.\log_e 5}{1}=\log_e 5$$

Sol. 20. (d)

$$\lim_{n\to\infty}\frac{1+2+3+.....+n}{1^2+2^2+3^3+.....+n^2}$$

$$\lim_{n\to\infty}\frac{\frac{n(n+1)}{2}}{\frac{n(n+1)(2n+1)}{6}}=\lim_{n\to\infty}\frac{3}{2n+1}=0$$

Sol. 21. (a)

$$\lim_{x\to 1}\frac{G(x)-f(1)}{x-1}$$

apply L. Hospital

$$\lim_{x\to 1}\frac{G(x)-f(1)}{x-1}=G'(x)$$

Now, $G'(x)=\frac{-2x}{2\sqrt{25-x^2}}$

$$G'(1)=\frac{-2}{2\sqrt{25-1}}=\frac{-1}{\sqrt{24}}=\frac{-1}{2\sqrt{6}}$$

Sol. 22. (b)

$$\lim_{x\to\infty}\left(\frac{2+x^2}{1+x}-Ax-B\right)=3$$

$$\lim_{x\to\infty}\left(\frac{2+x^2-Ax(1+x)-B(1+x)}{1+x}\right)=3$$

$$\lim_{x\to\infty}\left(\frac{x^2(1-A)-x(A+B)+2-B}{1+x}\right)=3$$

for definite result of this algebraic limit coefficient of x^2 should be zero.

$\Rightarrow 1-A=0$ so $A=1$

$-(A+B)=3$ so $B=-4$

Sol. 23. (c)

solution in above question.

Sol. 24. (d)

$$\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}$$

apply L. Hospital

$$\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}=f'(x)$$

Now, $f'(x)=\frac{-2x}{2\sqrt{25-x^2}}$

$$f'(1)=\frac{-2}{2\sqrt{25-1}}=\frac{-1}{\sqrt{24}}$$

Sol. 25. (d)

$$\lim_{x\to 2}\frac{\sin\left(e^{x-2}-1\right)}{\ln(x-1)}$$

apply L. Hospital

$$\lim_{x\to 2}\frac{\cos\left(e^{x-2}-1\right)e^{x-2}}{1/(x-1)}=1$$

Sol. 26. (b)

Given $f(x)=\frac{a^{[x]+x}-1}{[x]+x}$

$$\lim_{x\to 0^+}f(x)=\lim_{x\to 0^+}\frac{a^{[x]+x}-1}{[x]+x}=\lim_{h\to 0}\frac{a^{[0+h]+(0+h)}-1}{[0+h]+(0+h)}$$

$$\lim_{h\to 0}\frac{a^{[h]+(h)}-1}{[h]+(h)}=\lim_{h\to 0}\frac{a^{(h)}-1}{(h)}=\log_e a$$

Sol. 27. (c)

$$\lim_{x\to 0^-}f(x)=\lim_{x\to 0^-}\frac{a^{[x]+x}-1}{[x]+x}=\lim_{h\to 0}\frac{a^{[0-h]+(0-h)}-1}{[0-h]+(0-h)}$$

$$\lim_{h\to 0}\frac{a^{-1+(h)}-1}{-1+(h)}=\frac{a^{-1}-1}{-1}=1-a^{-1}$$

Sol. 28. (a)

$$\lim_{x\to 0}e^{-\frac{1}{x^2}}=e^{-\frac{1}{0}}=0$$

Sol. 29. (a)

$$\lim_{x\to 0}\phi(x)=a^2$$

$$\lim_{x\to 0}\phi\left(\frac{x}{a}\right)=a^2$$

because function value is constant.

Sol. 30. (b)

$$\lim_{x\to 0}\frac{e^x-(1+x)}{x^2}$$

apply L. hospital rule

$$\lim_{x\to 0}\frac{e^x-1}{2x}$$

again apply L. hospital rule

$$\lim_{x\to 0}\frac{e^x}{2}=\frac{1}{2}$$

Sol. 31. (c)

$$\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}$$

apply L. Hospital

$$\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}=f'(x)$$

Now, $f'(x)=\frac{-2x}{2\sqrt{9-x^2}}$

$$f'(1)=\frac{-2}{2\sqrt{9-1}}=\frac{-1}{2\sqrt{2}}$$

Sol. 32. (a)

If $\lim_{x\to a}f(x)$ and $\lim_{x\to a}g(x)$ both exits, then $\lim_{x\to a}\{f(x)g(x)\}$ exists. But if $\lim_{x\to a}\{f(x)g(x)\}$ exists, then it is not necessary that both $\lim_{x\to a}f(x)$ and $\lim_{x\to a}g(x)$ exits.

Sol. 33. (c)

$$\lim_{x\to\frac{\pi}{2}}\frac{\sin x}{x}=l \text{ and } \lim_{x\to\infty}\frac{\cos x}{x}=m,$$

$$l=\lim_{x\to\frac{\pi}{2}}\frac{\sin x}{x}=\frac{1}{\frac{\pi}{2}}=\frac{2}{\pi},$$

$$m=\lim_{x\to\infty}\frac{\cos x}{x}=\frac{\text{value between -1 to 1}}{\infty}=0$$

Sol. 34. (d)

$$\lim_{h\to 0}\frac{\sqrt{2x+3h}-\sqrt{2x}}{2h}$$

rationalize the numerator

$$\lim_{h\to 0}\frac{\sqrt{2x+3h}-\sqrt{2x}}{2h}\times\frac{\sqrt{2x+3h}+\sqrt{2x}}{\sqrt{2x+3h}+\sqrt{2x}}$$

$$\lim_{h\to 0}\frac{2x+3h-2x}{2h\left(\sqrt{2x+3h}+\sqrt{2x}\right)}=\frac{3}{4\sqrt{2x}}$$

Sol. 35. (a)

$$\lim_{x\to 0}\frac{\tan x}{\sin 2x}\times\frac{2x}{2x}$$

$$=\frac{\tan x}{x}\times\frac{2x}{\sin 2x}\times\frac{1}{2}=\frac{1}{2}$$

Sol. 36. (d)

$$\lim_{x\to\frac{\pi}{6}}\frac{2\sin^2 x+\sin x-1}{2\sin^2 x-3\sin x+1}$$

$$\lim_{x\to\frac{\pi}{6}}\frac{(\sin x+1)(2\sin x-1)}{(\sin x-1)(2\sin x-1)}=\lim_{x\to\frac{\pi}{6}}\frac{(\sin x+1)}{(\sin x-1)}=-3$$

Sol. 37. (c)

$$\lim_{\theta\to 0}\frac{\sqrt{1-\cos\theta}}{\theta}=\lim_{\theta\to 0}\frac{\sqrt{2}\sin(\theta/2)}{\theta}$$

$$=\frac{\sqrt{2}}{2}=\frac{1}{\sqrt{2}}$$

Sol. 38. (a)

$$\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}$$

apply L. Hospital

$$\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}=f'(x)$$

Now, $f'(x)=\frac{-2x}{2\sqrt{25-x^2}}$

$$f'(1)=\frac{-2}{2\sqrt{25-1}}=\frac{-1}{\sqrt{24}}$$

Sol. 39. (c)

$$\lim_{x\to 0}\frac{1-\cos^3 4x}{x^2}$$

apply L. Hospital rule

$$\lim_{x\to 0}\frac{3\cos^2 4x.\text{Sin}\,4x.4}{2x}$$

$$\lim_{x\to 0}\frac{3\cos^2 4x.\text{Sin}\,4x.4}{4x}\times 2$$

$$\left(\because \lim_{x\to 0}\frac{\text{Sin}4x}{4x}=1\right)$$

$=3(4)(2)=24$

Sol. 40. (b)

$$\lim_{x\to 0}\frac{\sin x^0}{\tan 3x^0}\times\frac{3x^0}{3x^0}$$

$$=\lim_{x\to 0}\frac{\sin x^0}{x^0}\times\frac{3x^0}{\tan 3x^0}\times\frac{1}{3}=\frac{1}{3}$$

Sol. 41. (a)

$$\lim_{x\to0}\frac{3^x+3^{-x}-2}{x}$$
apply L. hospital
$$\lim_{x\to0}\frac{3^x\log3-3^{-x}\log3}{1}=0$$
Sol. 42. (d)
$$\lim_{x\to1}\frac{x+x^2+x^3-3}{x-1}$$
apply L. hospital
$$\lim_{x\to1}\frac{1+2x+3x^2}{1}=6$$
Sol. 43. (c)
$$\lim_{x\to1}\frac{x^4-1}{x-1}=\lim_{x\to k}\frac{x^3-k^3}{x^2-k^2}$$
apply L. hospital
$$\lim_{x\to1}\frac{x^4-1}{x-1}=\lim_{x\to1}\frac{4x^3}{1}=4$$
and
$$\lim_{x\to k}\frac{x^3-k^3}{x^2-k^2}=\lim_{x\to k}\frac{3x^2}{2x}=\lim_{x\to k}\frac{3x}{2}=\frac{3k}{2}$$
$$\frac{3k}{2}=4\Rightarrow k=\frac{8}{3}$$
Sol. 44. (a)
$$\lim_{x\to0}\frac{\sin x\log(1-x)}{x^2}$$
$$\lim_{x\to0}\frac{\sin x}{x}.\frac{\log(1-x)}{x}=1(-1)=-1$$
Sol. 45. (c)
$$\lim_{x\to-1}\frac{f(x)}{x^2-1}=-\frac{3}{2}$$
by applying limit it gives zero in denominator
but answer is a real number given
so numerator will be also zero by applying limit.
$$\lim_{x\to-1}f(x)=f(-1)=0$$
Sol. 46. (c)
$$\lim_{x\to a}\frac{a^x-x^a}{x^a-a^a}=-1$$
apply L. hospital
$$\lim_{x\to a}\frac{a^x\log a-ax^{a-1}}{ax^{a-1}}=-1$$
$$\frac{a^a\log a-a.a^{a-1}}{a.a^{a-1}}=-1$$
$$\frac{a^a\log a-a.}{a}=-1$$
$$a^a\log a-a=-a$$
$$a^a\log a=0$$
$$a=1$$
Sol. 47. (b)
$$\lim_{x\to-1}\frac{x^3+x^2}{x^2+3x+2}$$
apply L. hospital
$$\lim_{x\to-1}\frac{3x^2+2x}{2x+3}=\frac{3-2}{-2+3}=1$$
Sol. 48. (c)
$$\lim_{n\to\infty}\frac{a^n+b^n}{a^n-b^n}$$ a > b > 1 as a > b
,
$$\Rightarrow\frac{b}{a}<1$$
So $$\lim_{n\to\infty}\frac{a^n\left[1+\left(\frac{b}{a}\right)^n\right]}{a^n\left[1-\left(\frac{b}{a}\right)^n\right]}$$
As $n\to\infty$ and $\frac{b}{a}<1$ So $\left(\frac{b}{a}\right)^n=0$

So $$\lim_{n\to\infty}\frac{\left[1+\left(\frac{b}{a}\right)^n\right]}{\left[1-\left(\frac{b}{a}\right)^n\right]}$$
$=\frac{1}{1}$ =1
,
Sol. 49. (d)
$$f(x)=\begin{cases}1+\frac{x}{2k}; & 0<x<2\\ kx; & 2<x<4\end{cases}$$
given $\lim_{n\to2}f(x)$

as $\lim_{n\to2}f(x)$ exist

so $\lim_{n\to2^-}f(x)\lim_{x\leftarrow2^+}$ f(x), $1+\frac{2}{2k}=2k$

$1+\frac{1}{k}=2k$

$K+1=2k^2$
$2k^2-k-1=0$
$2k^2-2k+k-1=0$
$2k(k-1)(2k+1)=0$
$(k-1)(2k+1)=0$
$K=1$ or $k=-\frac{1}{2}$

Sol. 50. (c)

f(x) = $\frac{[x]}{|x|}$ $x\neq0$

$$f(x)=\begin{cases}0 & x<1\\ 1 & x=1\\ \frac{1}{x} & x>1\end{cases}$$
$\underset{x\to1^+}{lt}\frac{[x]}{|x|}$ [1 + h] = 1

Or $\underset{h\to0}{lt}$ f(1+h)

$\underset{h\to0}{lt}\frac{[1+h]}{|1+h|}=\frac{1}{1}is1$

Sol. 51. (a)

$\lim_{x\to0}x^3(\operatorname{cosec}x)^2$
$$=\lim_{x\to0}\frac{x^3}{\sin^2x}$$
$$=\lim_{x\to0}x.\frac{x^2}{\sin^2x}$$
$=0\times1=0$

Sol. 52. (c)
$$=\lim_{x\to1}\frac{x^3-1}{\sqrt{x}-1}$$
Using L.hospital rule
$$\Rightarrow=\lim_{x\to1}\frac{3x^2}{1/2\sqrt{x}}$$
$$\Rightarrow=\lim_{x\to1}6x^2\sqrt{x}=6$$
Sol. 53. (d)
$$\lim_{x\to0}\frac{x}{\sqrt{1-\cos4x}}$$
$$=\lim_{x\to0}\frac{x}{\sqrt{2\sin^22x}}=\lim_{x\to0}\frac{2x}{2\sqrt{2}\,|\sin2x|}$$
For R.H.L.
$$=\lim_{h\to0}\frac{2h}{2\sqrt{2}\sin2h}=\frac{1}{2\sqrt{2}}$$
For L.H.L.
$$=\lim_{h\to0}\frac{-2h}{2\sqrt{2}\sin2h}=-\frac{1}{2\sqrt{2}}$$
Limit does not exists.

Sol. 54. (a)
$$\lim_{x\to\frac{\pi}{2}}\frac{4x-2\pi}{\cos x}$$
Apply L hospital rule
$$\lim_{x\to\frac{\pi}{2}}\frac{4}{-\sin x}=-4$$
Sol. 55. (d)
$$f(x)=\frac{x^2+x+|x|}{x}$$
R.H.L. at x = 0
$$=\lim_{h\to0}\frac{h^2+h+h}{h}=\frac{h^2+2h}{h}=h+2=2$$
L.H.L. at x = 0
$$=\lim_{h\to0}\frac{h^2+h-h}{h}=\frac{h^2}{h}=h=0$$
$\lim_{x\to0}f(x)$ does not exist

Sol. 56. (c)
$$\lim_{h\to0}\frac{\sin^2(x+h)-\sin^2x}{h}$$
Apply L hospital rule
Diff. w.r.t. h
$$\lim_{h\to0}\frac{2\sin(x+h)\cos(x+h)}{1}$$
=2sinxcosx = sin2x

Sol. 57. (d)
$$\lim_{x\to5}\frac{5-x}{|x-5|}$$
R.H.L. at x = 5
$$\lim_{h\to0}\frac{5-(5+h)}{|5+h-5|}=\lim_{h\to0}\frac{-h}{h}=-1$$
L.H.L. at x = 5
$$\lim_{h\to0}\frac{5-(5-h)}{|5-h-5|}=\lim_{h\to0}\frac{h}{h}=1$$
Limit does not exist.

Sol. 58. (d)

$$\lim_{x\to 1}\frac{x^9-1}{|x^3-1|}$$
For R.H.L
$$\lim_{x\to 1}\frac{x^9-1}{x^3-1}$$
Apply L. hospital rule
$$\lim_{x\to 1}\frac{9x^8}{3x^2}=3$$
For L.H.L
$$\lim_{x\to 1}\frac{x^9-1}{-\left(x^3-1\right)}$$
Apply L. hospital rule
$$\lim_{x\to 1}\frac{9x^8}{-3x^2}=-3$$
Limit does not exists

Sol. 59. (b)
$$f(x)=\begin{vmatrix}\cos x & x & 1\\ 2\sin x & x^2 & 2x\\ \tan x & x & 1\end{vmatrix}$$
$$f(0)=\begin{vmatrix}1 & 0 & 1\\ 0 & 0 & 0\\ 0 & 0 & 1\end{vmatrix}=0$$

Sol. 60. (b)
$$\frac{f(x)}{x}=\begin{vmatrix}\cos x & 1 & 1\\ 2\sin x & x & 2x\\ \tan x & 1 & 1\end{vmatrix}$$
$$=\lim_{x\to 0}\frac{f(x)}{x}=\begin{vmatrix}1 & 1 & 1\\ 0 & 0 & 0\\ 0 & 1 & 1\end{vmatrix}=0$$

Sol. 61. (a)
$$\frac{f(x)}{x^2}=\begin{vmatrix}\cos x & 1 & 1\\ 2\frac{\sin x}{x} & 1 & 2\\ \tan x & 1 & 1\end{vmatrix}$$
$$=\lim_{x\to 0}\frac{f(x)}{x^2}=\begin{vmatrix}1 & 1 & 1\\ 2 & 1 & 2\\ 0 & 1 & 1\end{vmatrix}=-1$$

Sol. 62. (b)
$$\lim_{x\to 1}\frac{a^{x-1}+b^{x-1}-1}{2(x-1)}$$
Apply L. hospital rule
$$=\lim_{x\to 1}\frac{a^{x-1}\log a+b^{x-1}\log b}{2}$$
$$=\frac{\log a+\log b}{2}=\frac{1}{2}\log ab$$

Sol. 63. (a)
$$\lim_{x\to 1}f(x)^{\frac{1}{g(x)}}$$
$$y=\lim_{x\to 1}\left(\frac{a^{x-1}+b^{x-1}}{2}\right)^{\frac{1}{x-1}}$$
taking log both sides
$$\log y=\lim_{x\to 1}\frac{\log\left(\frac{a^{x-1}+b^{x-1}}{2}\right)}{x-1}$$
Apply L. hospital rule
$$\log y=\lim_{x\to 1}\frac{\left(\frac{2}{a^{x-1}+b^{x-1}}\right)\left(\frac{a^{x-1}\log a+b^{x-1}\log b}{2}\right)}{1}$$
$$\log y=\frac{\left(\frac{2}{1+1}\right)\left(\frac{\log a+\log b}{2}\right)}{1}=\frac{1}{2}\log ab$$
$$\log y=\log\sqrt{ab}$$
$$y=\sqrt{ab}$$

Sol. 64. (b)
$$h(x)=\frac{f(g(x))}{g(f(x))}=\frac{|[x]-1|}{[|x|]-1}$$
$$\lim_{x\to 0+}h(x) = \text{R.H.L at } x = 0$$
$$\lim_{h\to 0}\frac{|[h]-1|}{[|h|]-1}=\lim_{h\to 0}\frac{|0-1|}{[h]-1}=\frac{1}{0-1}=-1$$

Sol. 65. (a)
$$h(x)=\frac{f(g(x))}{g(f(x))}=\frac{|[x]-1|}{[|x|]-1}$$
$$\lim_{x\to 0+}h(x) = \text{:L.H.L at } x = 0$$
$$\lim_{h\to 0}\frac{|[-h]-1|}{[|-h|]-1}=\lim_{h\to 0}\frac{|-1-1|}{[h]-1}=\frac{2}{0-1}=-2$$

Sol. 66. (a)
$f(x) = |x| + 1$
$g(x) = [x] - 1$
$$h(x)=\frac{f(x)}{g(x)}$$
$$\lim_{x\to 0-}h(x)+\lim_{x\to 0+}h(x)$$
$$=\lim_{x\to 0^-}\frac{|x|+1}{[x]-1}+\lim_{x\to 0^+}\frac{|x|+1}{[x]-1}$$
$$=\lim_{h\to 0}\frac{|0-h|+1}{[0-h]-1}+\lim_{h\to 0}\frac{|0+h|+1}{[0+h]-1}$$
$$=\lim_{h\to 0}\frac{h+1}{-1-1}+\lim_{h\to 0}\frac{h+1}{0-1}$$
$$=\frac{1}{-2}+\frac{1}{-1}=-\frac{3}{2}$$

Sol. 67. (d)
$$\lim_{x\to 3}\frac{|x-3|}{x-3}$$
R.H.L. at x = 3
$$\lim_{h\to 0}\frac{|3+h-3|}{3+h-3}=\lim_{h\to 0}\frac{h}{h}=1$$
L.H.L. at x = 3
$$\lim_{h\to 0}\frac{|3-h-3|}{3-h-3}=\lim_{h\to 0}\frac{h}{-h}=-1$$
Limit does not exist.

Sol. 68. (b)
$$\lim_{\theta\to\frac{\pi}{2}}(\sec\theta-\tan\theta)$$
$$=\lim_{\theta\to\frac{\pi}{2}}\left(\frac{1-\sin\theta}{\cos\theta}\right)$$
apply L. Hospital rule
$$=\lim_{\theta\to\frac{\pi}{2}}\left(\frac{-\cos\theta}{-\sin\theta}\right)=0$$

Sol. 69. (b)
$$\lim_{x\to 0}\{f(x)g(x)\}$$
For R.H.L.
$$\lim_{h\to 0}\sin[h]\,|h|=\lim_{h\to 0}\sin 0\,|h|=0$$
For L.H.L.
$$\lim_{h\to 0}\sin[-h]\,|-h|=\lim_{h\to 0}\sin(-1)\,|-h|=0$$
So limit is 0.

Sol. 70. (d)
$$\lim_{x\to 0}\frac{f(x)}{g(x)}$$
For R.H.L.
$$\lim_{h\to 0}\frac{\sin[h]}{|h|}=\lim_{h\to 0}\frac{\sin 0}{|h|}=0$$
For L.H.L.
$$\lim_{h\to 0}\frac{\sin[-h]}{|-h|}=\lim_{h\to 0}\frac{\sin(-1)}{|-h|}=\lim_{h\to 0}\frac{-\sin(1)}{h}=-\infty$$
limit does not exists.

Sol. 71. (a)
$f(x) = x^2 - 1$
$fof(x) = (x^2 - 1)^2 - 1$
$fof(1) = (1 - 1)^2 - 1 = - 1$

Sol. 72. (c)
$f(x) = x^2 + 9$
$f(0) = 9$
$$\lim_{x\to 0}\frac{\sqrt{f(x)}-3}{\sqrt{f(x)+7}-4}$$
Apply L hospital
$$=\lim_{x\to 0}\frac{\frac{1}{2\sqrt{f(x)}}f'(x)}{\frac{1}{2\sqrt{f(x)+7}}f'(x)}=\frac{\frac{1}{2\sqrt{f(0)}}}{\frac{1}{2\sqrt{f(0)+7}}}=\frac{\frac{1}{\sqrt{9}}}{\frac{1}{\sqrt{16}}}=\frac{4}{3}$$

Sol. 73. (a)
L.H.L. at 0
$$\lim_{h\to 0}f(0-h)=\lim_{h\to 0}\frac{1-\cos 2(-h)}{(-h)^2}$$
$$\lim_{h\to 0}\frac{1-\cos 2(h)}{(h)^2}=\lim_{h\to 0}\frac{2\sin^2 h}{h^2}=2$$

Sol. 74. (d)
R.H.L. at 0
$$\lim_{h\to 0}f(0+h)$$
$$=\lim_{h\to 0}\frac{\sqrt{h}}{\sqrt{\left(16+\sqrt{h}\right)}-4}\times\frac{\sqrt{\left(16+\sqrt{h}\right)}+4}{\sqrt{\left(16+\sqrt{h}\right)}+4}$$
$$=\lim_{h\to 0}\frac{\sqrt{h}}{\sqrt{16+h-16}}\times\frac{\sqrt{\left(16+\sqrt{h}\right)}+4}{1}=8$$

Sol. 75. (a)
Left hand limit at x = – 1
$$\lim_{h\to 0}f(-1-h)=\lim_{h\to 0}(-1-h)|-1-h|$$
$$=(-1)|-1|=-1$$